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Started	Wednesday, 28 October 2026, 7:45 PM
Completed	Wednesday, 28 October 2026, 8:02 PM
Duration	16 mins 57 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(log n) ✓
c. O(1)
d. O👎

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. n ✓
c. 1
d. log n

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Hash search

c. Binary search

d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It works on unsorted lists

c. It can be applied only if the list is sorted ✓

d. It compares elements sequentially

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists

b. It is slower than linear search for small lists

c. It requires the list to be sorted ✓

d. It does not work with negative numbers

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is sorted ✓
c. When the list contains duplicate elements
d. When the list is very small

Question **8**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 94 and 99 ✗
c. 90 and 99
d. 89 and 94

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Sort the list
c. Divide the list into two equal parts
d. Compare the target element with the first element in the list

Question **10**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search starts over from the beginning
- c. The search continues until the end of the list
- d. The search backtracks to find duplicate elements

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n-1
- c. n/2
- d. log n

Question **12**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search works on unsorted lists
- c. Binary search can find multiple instances of the target element
- d. Binary search does not require dividing the list

Question **13**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Inserting elements into a list
- b. Sorting elements in a list
- c. Determining whether an element is present in a list ✓
- d. Deleting elements from a list

Question **14**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Linear
- c. Both (1) & (3)
- d. Hash

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The last element is returned
- c. An error is raised
- d. The first element is returned